Personal AI Collaboration in Changing Spaces

Shared understanding and revisable action in XR

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Research vision and motivating problem

I want to develop personal AI assistants that people can talk to while doing things: assistants that understand their surroundings, contribute useful actions, retain useful memories and develop a consistent way of interacting across activities. Speech, spatial information and manipulable models could give this collaboration a place in the world. My doctoral starting point is how a person and an assistant establish what they are trying to do together, make that understanding actionable, and revise it as circumstances change.

Consider preparing a small exhibition. I point and say, 'Turn these towards the entrance, but keep that work visible.' The assistant must identify the exhibits, interpret the viewing relationship and propose a suitable change. I may correct its target, demonstrate a different angle or change the visitor route after walking through the room. A useful collaborator should carry forward what we have agreed while helping revise what no longer works. This is a problem of continuing joint action as well as recognising an instruction.

I propose to investigate a shared action representation: a small, directly editable connection between the assistant's selected objects, the spatial relations the user wants, and its pending actions. The representation would let a person correct a destination, preserve a viewing relationship or replace a next step while continuing the task. The research asks when making these connections explicit improves collaboration, and when it adds unnecessary effort.

Research context and opportunity

GazePointAR grounds contextual references in gaze, pointing and conversation [1]. GesPrompt maps speech and co-speech gestures to spatial function calls [2], while VRCopilot supports layout co-creation using editable wireframes [3]. AgentHands produces interactive, spatially grounded hand gestures [4]. Eye2Eye already combines shared attention, revisable contextual memory and feedback-based repair [5]. These systems motivate investigating how users can shape the connection between an assistant's interpretation and its next action.

I will build on this work to study a user-editable representation of action dependencies: letting people directly revise the relations between current intent and pending executable actions, then retain or update those relations after a goal or scene change. Human collaborative tasks show how linguistic and gestural conventions develop through repeated interaction [6]. This motivates studying which agreements should persist, which need checking, and how a correction can affect the next action without forcing a complete restart.

Research questions

RQ1. What spatial references, goals and action dependencies do people need to establish and repair when collaborating with an AI assistant?

RQ2. How does a persistent, editable shared action representation affect successful collaboration and the effort of correcting misunderstandings?

RQ3. When goals or spatial conditions change, how can the pair update affected actions while preserving agreements that remain useful?

Interaction concept and implementation

The first testbed will be an immersive exhibition workspace with four to eight movable exhibits, display panels and a known entrance. Participants will arrange, inspect and revise a layout. The task uses viewpoint, occlusion and spatial relationships directly, while providing a bounded setting for developing a more general collaborator. My architectural and CAVE experience supplies a practical starting point; the research concerns coordination mechanisms that could later inform other spatial work.

A person will speak, select objects and demonstrate destinations or orientations. The assistant will connect relevant objects to a concise statement of the intended relation and a preview of the next action. For example, a visible viewing path can express 'keep this visible from the entrance'; a ghost orientation shows the proposed rotation. The user can select a different referent, drag the destination or revise the relation through speech. Only relevant parts will appear, with a compact task overview available on demand. Early design work will establish when ordinary conversation is enough.

I will build one Unity/OpenXR prototype, beginning with desktop debugging and then using laboratory headset and controller access. Existing speech transcription, tracking and a pretrained language or vision-language model will provide input and interpretation. A known scene graph, object IDs and a finite tool API will support move, rotate, group, explain and compare actions. Structured model output will link referents, desired relations, pending actions and dependencies. A validator will check object identity, supported operations and simple geometric constraints before producing a preview. Confirmed digital actions will execute through the API with undo; physical rearrangement in later studies will be performed by the participant.

The action state will distinguish user-confirmed agreements from model assumptions, and record the observation and scene version behind each pending action. Changes will mark affected dependencies for review. The assistant can ask for clarification or propose an update, which the user can inspect and modify. The first system will support speech and controller pointing; free-hand techniques will be selected if formative findings justify their effort. Body-centred interfaces such as Handows [7] and on-body disambiguation [8] provide relevant design precedents.

Study 1 Understanding how spatial collaboration develops

I will observe approximately 12-16 participants working in pairs on small layout and inspection tasks, including people with and without design training. Replay interviews will examine how they establish referents, explain desired effects, divide actions and repair an earlier interpretation. Analysis will retain ambiguous cases and distinguish target, goal, constraint, sequencing and execution problems. A small Wizard-of-Oz iteration within this formative phase will test candidate representations and corrections before model reliability dominates the interaction. Researcher-controlled behaviour will be disclosed and logged. The study will produce a design space of coordination and repair, representative change events and requirements for the prototype.

Study 2 Editing understanding while continuing action

A counterbalanced within-participant study will compare a conversational assistant with object highlighting, action previews and dialogue-based corrections to the current task state against the same assistant with the persistent shared action representation. Both will use the same XR platform, input modalities, model, scene information, action API, confirmation and undo. Both will permit corrections to targets, goals and constraints. The comparison estimates the value of the linked representation and correction workflow as a whole.

Matched tasks will include ordinary successful collaboration and controlled, reversible mistakes in a selected object, an intended relation or an inherited goal. The same error instances will be inserted into structured proposals at matched task opportunities in both conditions; natural model errors will be logged separately. This tests recovery from comparable misunderstandings alongside the natural errors reported for the running model. Approximately 24-30 participants is an initial recruitment range; pilot variability and a prespecified precision target will determine the final sample.

Study 2 Evaluation and analysis

The primary outcome will be successful repair before the first confirmed execution of the affected action: the mistaken interpretation is corrected and the proposed action meets the independently specified task requirement. Time to reach that state, clarification turns, unintended changes to agreed parts and final layout constraints will capture practical costs. Free-design tasks will use the participant's stated and revisable goal as the reference, separating aesthetic judgement from geometric criteria. Brief probes of what the assistant is about to do, workload and perceived agency will complement behaviour. Analysis will account for repeated observations within participants and tasks, with effect estimates and uncertainty.

Study 3 Updating a joint task across changing situations

I will extend the same prototype to a room-scale arrange-inspect-revise workflow. Changes will include a moved object, a revised visitor route or a shift from arrangement to rehearsal. A first physical transfer can use known objects and markers linked to a digital counterpart; this avoids making general-purpose scene reconstruction a prerequisite. A planning sample of 12-16 participants across two short visits will examine early reuse and continuation. The task and equipment will be finalised after the previous system is stable.

Both conditions will retain the same shared action interface, available observations, interaction history, model and manual editing controls. One will present an ordinary refreshed task summary and proposals; the other will explicitly link changed evidence to affected relations and actions, proposing a local revision while retaining still-valid agreements. Matched change opportunities will test the update protocol, with the first proposal after each registered change serving as the update checkpoint. The model will operate on actual observations and records, without an undisclosed operator correcting its decisions.

The main outcome will be incorrect carryover per assessable change opportunity: after the update checkpoint, an action presented as current and executable, or actually executed, still relies on an object, relation or goal invalidated by the recorded change. Actions marked as awaiting clarification will be counted separately. Resumption time, unnecessary replanning of agreed work, task quality and supervision effort will reveal trade-offs. Errors will be separated into missed observations, mistaken interpretation, stale dependencies and failed execution. Offline scenarios and a deployment pilot will first check valid actions, retained agreements and clarification behaviour. These visits examine early reuse; longer-term personalisation remains a further stage of the broader vision.

Contributions preparation and research fit

The programme aims to contribute an empirical account of spatial coordination and repair, an implemented mechanism connecting editable intent to action, and evidence about updating collaboration as activities change. This directly connects to the Situated Reality Lab's situational spatial intelligence, adaptive interfaces and embodied AI with shared agency [9]. My spatial-design and installation experience can ground realistic tasks; Game Fellows motivates the move from following behaviour to a partner that contributes to joint activity. My image-detection capstone adds experience in reproducible evaluation and analysing failures.

Year 1 will develop Unity/C# skills, complete the formative work and establish an end-to-end prototype. Year 2 will develop and evaluate the shared action mechanism. Year 3 will examine updating and physical transfer, followed by synthesis and necessary follow-up in Year 4. Existing models and tracking components will be reused. My browser explorations provide an initial interaction testbed; building and testing a headset prototype is my next technical step.

Ethics approval will precede recruitment. Recording will be consented and limited to the task area; participants can pause sensing, take breaks and withdraw. Model-bound inputs will exclude unrelated personal material, and any external processing will be disclosed. Interaction logs will record model, tracking and human-intervention failures separately.

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